

Assignment-2

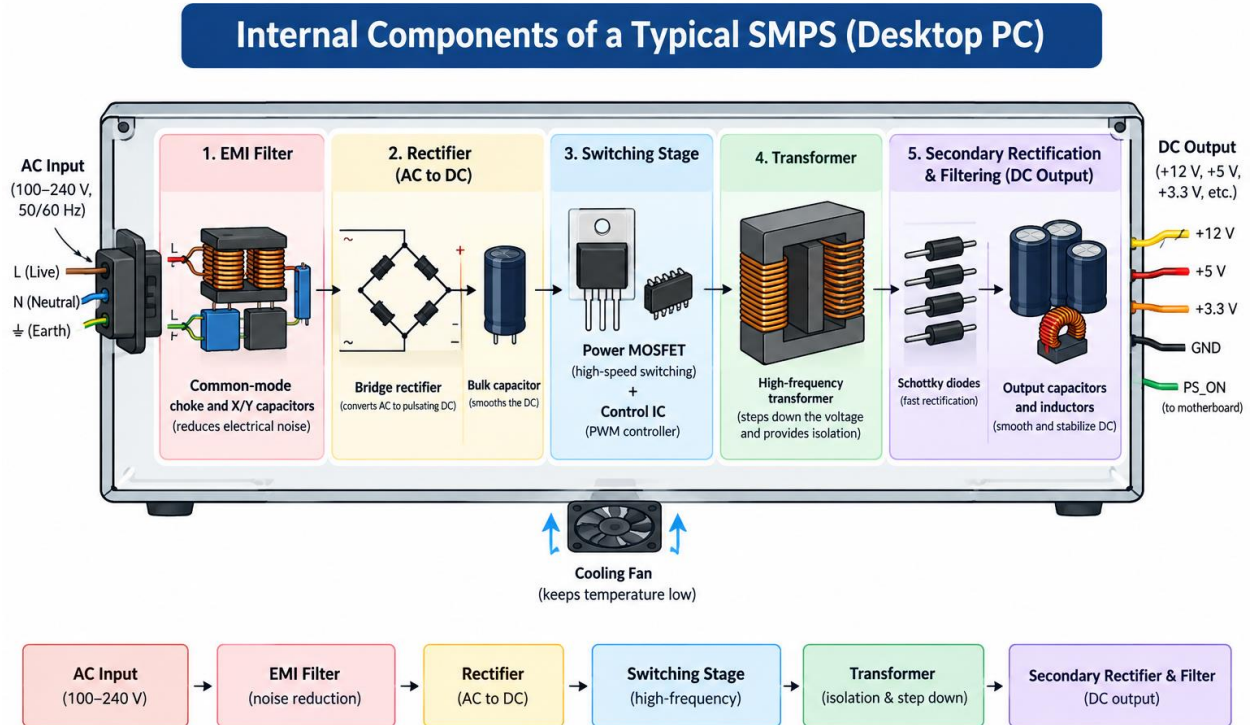
Day 2: Power Supply Systems Theory: Basics of SMPS and UPS systems.

Practical: identify power cables and connectors.

Task-1: - List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Type	Device	Why it needs this power supply
SMPS	Desktop Computer	Converts AC power into the different DC voltages required by the computer components.
SMPS	LED/LCD Monitor	Converts household AC power into the low-voltage DC power needed by the monitor.
SMPS	Wi-Fi Router	Provides the required low-voltage DC power efficiently from household AC power.
UPS	Desktop Computer	Provides temporary backup power during a power cut so work can be saved, and the computer can be shut down safely.
UPS	Wi-Fi Router	Keeps the router running during short power interruptions so the internet connection can remain available.

Task-2: - Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.



Task-3: Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Common Power Connectors in a PC

1. ATX 24-pin Motherboard Connector



Name:
ATX 24-pin
Motherboard Connector



Powers:

- Motherboard
- (provides main power to the system)



Safety Tip:
Always turn off the power supply and unplug the PC before connecting or disconnecting this connector.

2. SATA Power Connector



Name:
SATA Power Connector



Powers:

- SSDs
- HDDs
- Optical drives
- Some case fans / accessories



Safety Tip:
Hold the connector by the plastic housing, not the wires, when plugging or unplugging.

3. Molex Connector



Name:
Molex Connector
(4-pin)



Powers:

- Older HDDs
- Optical drives
- Case fans
- Some legacy devices



Safety Tip:
Do not force the connector. Check the correct orientation and make sure it is fully seated before use.

4. PCIe 6/8-pin Connector



Name:
PCIe 6/8-pin Connector
(6+2 pin)



Powers:

- Graphics cards (GPU)
- High-power devices (e.g., some expansion cards)



Safety Tip:
Make sure the connector is fully clicked in and locked. Do not bend or pull the cable with excessive force.



Always ensure your PC is powered off and unplugged before connecting or disconnecting any power connector.

Task-4: Imagine setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step 1: Install the SMPS

Place the SMPS/PSU in the PC cabinet and secure it using the appropriate screws.

Step 2: Connect the Motherboard

- Take the **24-pin ATX motherboard cable** from the SMPS.
- Connect it to the 24-pin power socket on the motherboard.
- If the motherboard/CPU requires it, connect to the **4/8-pin CPU/EPS cable** near the CPU socket.

Step 3: Connect the Graphics Card

- Identify the **PCIe 6-pin or 8-pin power cable** from the SMPS.
- Connect it to the power socket(s) on the graphics card.
- Some high-end GPUs may use newer dedicated power connectors, so use the cable specified for that GPU and PSU.

Step 4: Connect Storage Devices

- For a **SATA SSD or HDD**, connect a **SATA power cable** from the SMPS to the drive.
- Connect a separate **SATA data cable** from the drive to the motherboard.
- An **M.2 NVMe SSD** normally does not require a separate PSU power cable; it receives power through its motherboard M.2 slot.

Step 5: Check All Connections

Make sure every connector is fully inserted and properly aligned. **Never force a connector.**

Step 6: Final Check and Power On

Check that cables are away from fans and that no loose connectors are touching components. Connect the AC power cable to the PSU, switch on the PSU, and then start the PC.